Networking Nature: Early Victorian Science and Politics in the Mass Press

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In my short paper, *Networking Nature: Early Victorian Science and Politics in the Mass Press*, I argue that natural science discourse served as political rhetoric in mass produced British periodicals from 1826 to 1848. Due to the emergent mass press, an era of increased popularization of science and scientific knowledge production became an era of increased political action, advocacy, and reform. In this historical period, from the adoption of the mechanized press to the end of the working class Chartist movement, early forms of open science were co-opted in the advocacy of a more open politics, as working class activists, feminist organizations, and moderate reformers used scientific knowledge to argue for greater representation and power in government. While various political groups engaged in this discourse, I argue that elite publishers and political authorities successfully used natural science discourse, with its veneer of respectable rationality, to moderate political radicalism in British society leading into the Victorian era.

I train and use Large Language Models to detect the discussion of natural science in large amounts of text data from the different political papers *Blackwood's Edinburgh Magazine*, *The Poor Men's Guardian*, and *The Isis*. Close readings of detected natural science discourse and attention to the materiality and format of their placement within periodicals will show how different publications used natural science to advance their political goals. Digital methods reveal the incorporation of natural science discourse into popular periodicals in a network of reprinting, while close reading and book history methods show how natural science texts were juxtaposed, combined, and recontextualized to produce political meanings. As the invocation of natural science within the popular press was an important mechanism of political debate of the time, this study shows how popular and participatory science can affect political culture and ideological debate.

Scholars of science and print culture have used book history methods and historical research to examine the audience reception, publishing processes, and authorial relationships that disseminate expert science to a mass reading audience. Authors like Jonathan Topham in *Reading the Book of Nature* and James Secord in *Victorian Sensation* focus on elite publishers and sensational texts. These studies rely on diaries, letters, and author networks to reconstruct reading practices and audience reception to understand the popularization of science before the work of Charles Darwin, but do not fully delve into the massive archive of political periodicals and how their discussion of science influenced larger cultural struggles for economic, political, and gender equality. Digital methods solve these limitations of scale, identifying reprinted discourse and its periodical context within a large digital archive. I draw on James Secord's idea of detecting culturally significant "traces" of scientific discussion with an archive, and I will rely on patterns of periodical discourse, not reader responses (2000, p. 5) . I combine these ideas with Ryan Cordell's methods of detecting and analyzing "viral texts" (2015, p. 421). I will apply coding techniques to detect significant scientific discussions and popular text reprints within a network and perform close readings of passages within their page format and context, moving "between macro- and micro-views" of the text (Cordell, 2015, p. 419). Using accessible digital archives, this study provides opportunities for close readings, moving between large network level views of discourse and attention to text as it appears and exists with the heterogenous context of the periodical page and issue.

I use the archive of the Hathi Trust Digital Library to access both text data and scanned page images of the periodicals, *Blackwood's Edinburgh Magazine*, *The Poor Man's Guardian*, and *The Isis* to conduct this study. These magazines, representing conservative, working class, and feminist opinion, show a range of how scientific discourse existed in the periodical network and shaped political advocacy. I train a large language model on scientific text data from these magazines as well as *The Penny Magazine*, a knowledge periodical of the era. This model will then be used to detect discussions of natural science in these three periodicals. Text reuse detection algorithms also indicate where identical passages appear across publications. Digital analysis of text data will identify significant discussion of science, while archival reading and attention to print material will determine the political use and meaning of this discussion. This study is a work in progress, serving as a smaller experiment exploring these three journals as I begin my larger *Networking Nature*

project focused on scientific discourse within the mass press of the early Victorian era. These methods reveal how science is integrated into a larger, diverse textual network and acts as both a form of new knowledge formation and a rhetorical strategy for political actors.

These three magazines, *Blackwood's Edinburgh Magazine*, *The Poor Man's Guardian*, and *The Isis* trace important conversations about creation and evolution and reflect diverse perspectives on government power. All three publications discuss the significance of expanded access to information, education, and scientific learning. However, *Blackwood's Edinburgh Magazine* uses ideas of evolution to rationalize existing power structures and argue for incremental parliamentary reforms, while *The Poor Man's Guardian* and *The Isis* uses discussion of creation and evolution to argue for political equality and greater representation in parliament for working class men and women. This study of the emergent political press from 1826 to 1848 shows the significance of popular cultures of natural science, as well as the textual influences that led to the rise and decline of British political radicalism in this era. My study intervenes to show that the noted tendencies towards secularism and moderate reform of the early Victorian era are influenced by the creation and use of natural science texts. The project's methods from both digital humanities and book history will serve as an example in studying larger information systems as important factors in cultural and political change.

Bibliography

Cordell, Ryan (2015): "Reprinting, Circulation, and the Network Author in Antebellum Newspapers", in: *American Literary History* 27, 3: 417–445. https://doi.org/10.1093/alh/ajv028 [16.04.2025].

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